Which ITIL practice area aligns IT services with business objectives?
A. Event Management
B. Service Strategy
C. Release Management
D. Service Desk
ANSWER: B

Which traditional ITIL v3 provider type is embedded within a business unit?
A. Type II
B. Type III
C. Type I
D. Public provider
ANSWER: C

Which provider type consolidates services for multiple business units in one organization?
A. Type II
B. Type III
C. External-only provider
D. Type I
ANSWER: A

Which provider type delivers services to external customers?
A. Type I
B. Type III
C. Type II
D. Internal shared unit
ANSWER: B

Which process helps evaluate services being considered, developed, offered, or retired?
A. Incident Management
B. Service Portfolio Management
C. Event Management
D. Release Management
ANSWER: B

Which Service Strategy process focuses on budgeting, accounting, and charging where applicable?
A. Knowledge Management
B. Change Management
C. Financial Management for IT Services
D. Capacity Management
ANSWER: C

Which process helps understand and influence demand for IT services?
A. Problem Management
B. Deployment Management
C. Access Management
D. Demand Management
ANSWER: D

A company compares business value, risk, total cost, and strategic fit before choosing a managed service. Which approach is this?
A. Strategic service evaluation
B. Routine deployment
C. Event correlation
D. Incident categorization
ANSWER: A

Why is service management considered a strategic asset?
A. It removes the need for business planning
B. It helps deliver value and achieve business outcomes
C. It guarantees zero operating cost
D. It replaces technical staff
ANSWER: B

Which information is most useful when deciding whether to retire a low-value service?
A. Only the server's brand
B. Only the number of configuration files
C. Demand, cost, risk, and business dependencies
D. Only the service's original launch date

ANSWER: C

What is a primary purpose of Service Design?
A. Close every incident
B. Archive all retired hardware
C. Deploy every release immediately
D. Design services and their supporting components to meet requirements
ANSWER: D

Which document provides details needed to develop and transition a new or changed service?
A. Service Design Package
B. Event log only
C. Incident record
D. Release note only
ANSWER: A

Which process maintains information about available services and their offerings?
A. Problem Management
B. Service Catalog Management
C. Event Management
D. Release Management
ANSWER: B

Which process defines, agrees, monitors, and reports service-level targets?
A. Financial Management
B. Knowledge Management
C. Service Level Management
D. Deployment Management
ANSWER: C

Which management area ensures service resources can meet agreed workload and performance needs?
A. Incident Management
B. Change Evaluation only
C. Service Desk Management
D. Capacity Management
ANSWER: D

Which capability plans how an IT service will continue or recover after a major disruption?
A. IT Service Continuity Management
B. Event Fulfillment
C. Service Catalog Management
D. Release Packaging
ANSWER: A

When designing information security, what should be considered?
A. Only screen resolution
B. Confidentiality, integrity, availability, and applicable requirements
C. Only application color schemes
D. Only hardware purchase date
ANSWER: B

Why should service architecture account for future business requirements?
A. To prevent all future changes
B. To eliminate documentation
C. To make the service adaptable as needs change
D. To avoid capacity planning
ANSWER: C

A service has no agreed availability target. Which process should establish and manage service-level agreements?
A. Problem Management
B. Asset Management
C. Event Management
D. Service Level Management
ANSWER: D

A design review finds that expected user growth was ignored. Which design area is most directly affected?
A. Capacity planning

B. Service catalog naming
C. Knowledge transfer only
D. Incident prioritization only
ANSWER: A

Which process maintains information about configuration items and their relationships?
A. Demand Management
B. Service Asset and Configuration Management
C. Service Catalog Management
D. Financial Management
ANSWER: B

Which activity coordinates resources and planning for service transition?
A. Event Fulfillment
B. Demand Management
C. Transition Planning and Support
D. Capacity Reporting
ANSWER: C

Which process controls the building, testing, and deployment of releases?
A. Service Level Management
B. Financial Management
C. Demand Management
D. Release and Deployment Management
ANSWER: D

Which process assesses and controls changes to services while managing risk?
A. Change Management
B. Service Catalog Management
C. Event Management
D. Demand Management
ANSWER: A

Which process helps ensure useful knowledge is available to the people who need it?
A. Capacity Management
B. Knowledge Management
C. Financial Management
D. Release Packaging
ANSWER: B

Which Service Operation process detects and interprets events from infrastructure and services?
A. Problem Management
B. Demand Management
C. Event Management
D. Service Portfolio Management
ANSWER: C

Which process restores normal service operation after an interruption?
A. Problem Management
B. Change Management
C. Service Portfolio Management
D. Incident Management
ANSWER: D

Which process investigates underlying causes of incidents to reduce recurrence?
A. Problem Management
B. Event Management
C. Incident Management
D. Service Catalog Management
ANSWER: A

Which function provides a central point of contact for users seeking help with IT services?
A. Release Management
B. Service Desk
C. Financial Management
D. Capacity Management
ANSWER: B

A service is reliable but unusually expensive to operate. Which operational consideration best

describes this challenge?
A. Cloning a virtual machine
B. Creating a service catalog entry only
C. Balancing conflicting goals
D. Performing a Git commit
ANSWER: C

What is the main aim of Continual Service Improvement (CSI)?
A. Freeze all service processes permanently
B. Replace incident records with emails
C. Prevent staff training
D. Identify and implement improvements to services and processes
ANSWER: D

Why are training and awareness activities included in improvement work?
A. To help people understand and perform changed practices
B. To avoid measuring results
C. To eliminate ownership
D. To replace service monitoring
ANSWER: A

Why should improvement activities have an assigned owner?
A. To ensure nobody makes decisions
B. To establish accountability for progress and outcomes
C. To avoid documenting tasks
D. To remove the need for review
ANSWER: B

What is the purpose of scheduling CSI activities?
A. To prevent any recurring review
B. To bypass prioritization
C. To coordinate improvement work and follow-up
D. To replace service-level targets
ANSWER: C

An improvement initiative has no person responsible for it. What is the clearest governance gap?
A. The service has too much capacity
B. The network uses IP
C. The service catalog is searchable
D. Ownership is not assigned
ANSWER: D

Which evidence is most useful when deciding whether an improvement worked?
A. Measures collected before and after the change
B. The project title alone
C. A single unsupported opinion
D. The number of meetings only
ANSWER: A

What should follow identification of a worthwhile improvement?
A. Close the issue without action
B. Plan, assign responsibility, implement, and review results
C. Remove all performance measures
D. Skip communication with affected staff
ANSWER: B

Why should improvement roles be clearly defined?
A. To make accountability impossible
B. To stop collaboration
C. To clarify responsibilities and decision-making
D. To eliminate training
ANSWER: C

A team changes a process but never checks its impact. Which CSI step is missing?
A. Naming the process
B. Opening a calendar
C. Assigning an unrelated asset tag
D. Reviewing and measuring the result
ANSWER: D

Which approach best supports ongoing service improvement?
A. Regularly review performance and act on evidence
B. Avoid recording lessons learned
C. Improve only after a major failure
D. Make changes without owners
ANSWER: A

Which factor is essential when planning data center physical space?
A. Only the color of racks
B. Equipment footprint plus service and maintenance clearance
C. Only the number of office chairs
D. Only the server operating-system name
ANSWER: B

Which facility system removes heat generated by IT equipment?
A. DNS
B. Git
C. Cooling and HVAC
D. IAM
ANSWER: C

Why must data center power requirements be calculated?
A. To determine domain names
B. To select a source-code branch
C. To create database indexes
D. To provide adequate capacity and support reliable operation
ANSWER: D

Which consideration is important when selecting a data center geographic location?
A. Power, connectivity, risk, and operational requirements
B. Only the rack color
C. Only proximity to a restaurant
D. Only the server hostname
ANSWER: A

What is a key purpose of modular cabling design?
A. Encrypt every application automatically
B. Organize structured connectivity and simplify moves or expansion
C. Replace cooling systems
D. Allocate cloud IAM roles
ANSWER: B

What does an ISP WAN link commonly provide to a data center?
A. Local disk snapshots only
B. Physical door access
C. Connectivity between the data center and external networks or sites
D. Operating-system patch management
ANSWER: C

What is a Network Operations Center (NOC) primarily used for?
A. Manufacturing network cables
B. Writing all application source code
C. Issuing employee payroll
D. Monitoring and coordinating network operations
ANSWER: D

Which is an example of physical data center security?
A. Controlled entry using badges and access procedures
B. A firewall rule alone
C. A database query filter
D. A Git commit signature
ANSWER: A

Which is an example of logical security in a data center?
A. Rack floor loading
B. Access controls and network security rules
C. HVAC duct placement
D. Cleaning a raised floor

ANSWER: B

Why is unoccupied space important in data center planning?
A. It guarantees that no cooling is needed
B. It replaces network bandwidth
C. It supports airflow, access, maintenance, and future expansion
D. It removes the need for fire protection
ANSWER: C

What does hardware virtualization allow a hypervisor to do?
A. Translate domain names
B. Manage service-level agreements
C. Publish Git commits
D. Run virtual machines using abstracted physical resources
ANSWER: D

Where does a Type 1 hypervisor generally run?
A. Directly on physical hardware
B. Only inside a browser tab
C. Only as an application inside a host operating system
D. Only on a network router
ANSWER: A

Where does a Type 2 hypervisor generally run?
A. Directly as the physical machine firmware only
B. As software hosted by a conventional operating system
C. Only inside a container image
D. Only on a storage array
ANSWER: B

What is para-virtualization associated with?
A. Running without any virtualization layer
B. Encrypting network traffic by default
C. Guest operating systems cooperating with the hypervisor through defined interfaces
D. Replicating a database table
ANSWER: C

What is a virtual machine snapshot primarily used for?
A. Replacing every backup strategy
B. Assigning a public IP automatically
C. Creating a Git remote
D. Capturing a point-in-time VM state for later reversion or reference
ANSWER: D

What is the usual purpose of cloning a virtual machine?
A. Creating another VM based on an existing VM
B. Rotating application logs
C. Measuring network latency only
D. Registering a DNS domain
ANSWER: A

What is a VM template?
A. A physical power distribution unit
B. A reusable baseline from which virtual machines can be deployed
C. A service-level agreement
D. A network packet capture
ANSWER: B

A lab needs isolated operating systems on one physical server. Which technology best fits?
A. Service catalog management
B. DNS caching
C. Virtualization
D. Source control only
ANSWER: C

Which statement about snapshots is most accurate?
A. They always protect against loss of the entire datastore
B. They eliminate the need to test recovery
C. They are the same as a new physical server

D. They are useful recovery points but should not automatically replace independent backups
ANSWER: D

An administrator wants a VM platform that runs directly on server hardware. Which type should be selected?
A. Type 1 hypervisor
B. Web browser
C. Type 2 hypervisor
D. Container registry
ANSWER: A

What is a Storage Area Network (SAN) designed to provide?
A. Domain-name resolution
B. Block-level storage access over a dedicated or shared storage network
C. Source-code collaboration
D. User-interface rendering
ANSWER: B

Which platform is named in the syllabus for SAN configuration?
A. Jenkins
B. Puppet
C. FreeNAS
D. Prometheus
ANSWER: C

What is a major benefit of using a SAN for high availability?
A. It removes the need for network connectivity
B. It guarantees that applications contain no bugs
C. It replaces identity management
D. Storage can be designed with redundancy and shared access for supported systems
ANSWER: D

Which filesystem and volume-management technology is explicitly listed for configuration?
A. ZFS
B. FAT12
C. NTFS only
D. ISO 9660
ANSWER: A

What does IP-based storage communication use to transport storage traffic?
A. Only USB cables
B. IP networks
C. Only serial console ports
D. Only Bluetooth
ANSWER: B

A server needs access to remote block storage rather than a shared folder protocol. Which architecture is most relevant?
A. DNS
B. Git repository
C. SAN
D. NOC dashboard
ANSWER: C

Why should storage redundancy be considered in a high-availability design?
A. To make every storage failure impossible
B. To eliminate access controls
C. To avoid monitoring capacity
D. To reduce the effect of certain component failures
ANSWER: D

What should an administrator verify before presenting a ZFS volume to a host?
A. Pool and volume configuration, access settings, and network connectivity
B. Only Git username
C. Only the desktop wallpaper
D. Only the public website title
ANSWER: A

Which statement best distinguishes a SAN from ordinary file sharing?

A. A SAN is always a web browser
B. A SAN commonly presents block storage, while file sharing presents files through a file protocol
C. A SAN only stores DNS records
D. A SAN is a source-code language
ANSWER: B

A host cannot reach IP-based storage. Which initial check is most relevant?
A. Change the Git commit message
B. Reinstall the web browser
C. Verify addressing, routing, network reachability, and storage service configuration
D. Modify the service catalog description
ANSWER: C

What does IaaS generally provide?
A. A complete end-user application only
B. Only source-code review
C. Only a physical office
D. Virtualized compute, storage, and networking resources
ANSWER: D

Which cloud service model provides a managed platform for deploying applications?
A. PaaS
B. SaaS
C. IaaS
D. NOC
ANSWER: A

Which model typically delivers a ready-to-use application to end users?
A. IaaS
B. SaaS
C. PaaS
D. SAN
ANSWER: B

Which deployment model uses infrastructure intended for a single organization?
A. Public cloud
B. Hybrid cloud only
C. Private cloud
D. Community Wi-Fi
ANSWER: C

Which deployment model combines public and private cloud environments?
A. Single-tenant on-premises only
B. Type 1 hypervisor
C. Local Git
D. Hybrid cloud
ANSWER: D

In the cloud SPI model, what does the letter P represent?
A. Platform as a Service
B. Power Infrastructure
C. Private Internet
D. Packet Inspection
ANSWER: A

What does IAM primarily manage in cloud environments?
A. HVAC temperature
B. Identities, authentication, and permissions
C. Docker image compression
D. Physical rack dimensions
ANSWER: B

What does a cloud Service Level Agreement (SLA) typically define?
A. Every application source-code line
B. A user's local wallpaper
C. Agreed service commitments and measurable targets
D. A VM snapshot's filename only
ANSWER: C

A workload needs an event-triggered function without managing a long-running server. Which listed AWS service category best matches this pattern?
A. Object storage only
B. A physical NOC
C. A Git commit
D. Serverless compute such as Lambda
ANSWER: D

When integrating with a cloud API, what is a sound security practice?
A. Use least-privilege credentials and protect secrets
B. Disable authentication
C. Embed unrestricted credentials in public source code
D. Share one administrator key publicly
ANSWER: A

What is Chef primarily used for in infrastructure automation?
A. Collecting metrics only
B. Defining and applying desired system configurations
C. Managing DNS records only
D. Editing images in a graphics editor
ANSWER: B

In Chef, what is a recipe commonly used to define?
A. A physical rack layout
B. A cloud billing invoice
C. Configuration resources and actions
D. A GitHub issue label
ANSWER: C

What is a Chef cookbook?
A. A storage protocol
B. A type of network switch
C. A Jenkins build artifact only
D. A collection of recipes, attributes, files, templates, and related configuration content
ANSWER: D

In a typical Chef architecture, which component stores cookbooks and policy information for managed nodes?
A. Chef Server
B. Docker Hub
C. Node Exporter
D. Git shell prompt
ANSWER: A

What is a Chef node?
A. A service-level target
B. A system managed by Chef
C. A storage volume format
D. A Git tag only
ANSWER: B

What is the purpose of installing a Chef client on a managed node in a traditional Chef setup?
A. To provide public DNS resolution
B. To replace the operating system kernel
C. To retrieve applicable policy and converge the node configuration
D. To export a Docker image
ANSWER: C

What does configuration convergence mean in infrastructure automation?
A. Randomly changing settings each run
B. Deleting all configuration files
C. Disabling all services
D. Bringing a system toward the declared desired state
ANSWER: D

A team must configure the same package and service settings across many servers. Which Chef capability is most relevant?
A. Reusable automated configuration

B. DNS reverse lookup
C. Manual one-time editing only
D. Spreadsheet formatting
ANSWER: A

Why should Chef resources be declared in a repeatable manner?
A. To ensure each run introduces unrelated changes
B. To support predictable configuration enforcement across runs
C. To disable auditability
D. To make node identity unnecessary
ANSWER: B

Before deploying a Chef cookbook to production, what is a sensible practice?
A. Run unreviewed changes on every node immediately
B. Remove all backups
C. Test the cookbook and validate its effect in a controlled environment
D. Publish private credentials in the cookbook
ANSWER: C

What is Puppet primarily used for?
A. Visualizing metrics only
B. Building container images only
C. Managing IT service finances
D. Automating and enforcing system configuration
ANSWER: D

Which language-like structure is central to describing desired configuration in Puppet?
A. Puppet manifests
B. SQL result sets
C. HTML forms
D. Packet capture filters
ANSWER: A

What is a Puppet module?
A. A type of WAN link
B. A reusable structure containing manifests and related files
C. A physical cooling unit
D. A cloud billing account
ANSWER: B

In a common Puppet architecture, what is the role of the agent?
A. Serve as the public container registry
B. Replace all network routers
C. Request and apply configuration for its managed node
D. Act only as a browser
ANSWER: C

What is a Puppet catalog?
A. A list of customer invoices
B. A Docker image archive
C. A DNS zone file only
D. A compiled description of resources and their desired relationships for a node
ANSWER: D

What is the role of a Puppet server in a typical agent-server deployment?
A. Compile and provide configuration catalogs to agents
B. Host only Git commits
C. Monitor CPU using Windows Exporter
D. Provide physical power to racks
ANSWER: A

What is a key benefit of declarative configuration management?
A. It guarantees no failures can occur
B. It describes desired state rather than requiring every low-level step to be scripted
C. It removes the need for credentials
D. It prevents reuse
ANSWER: B

A configuration must be applied consistently to dozens of Linux systems. Which Puppet use is most

appropriate?
A. Manually edit each host with no record
B. Use a DNS query as the only configuration method
C. Manage package, file, and service state through manifests
D. Create a separate physical server for every setting
ANSWER: C

Which practice reduces risk when introducing a new Puppet module?
A. Disable all monitoring
B. Store secrets in public repositories
C. Apply it without reviewing changes
D. Test it on representative non-production nodes first
ANSWER: D

What should be investigated if a Puppet-managed service repeatedly returns to an unexpected state?
A. The declared resources, dependencies, and catalog applied to the node
B. Only the GitHub profile picture
C. Only the monitor's brightness
D. The office seating plan
ANSWER: A

What type of monitoring data does Prometheus primarily collect and store?
A. Only source-code diffs
B. Time-series metrics
C. Only word-processing documents
D. Only physical access cards
ANSWER: B

Which component commonly exposes Linux host metrics for Prometheus?
A. Windows Exporter
B. Docker Compose
C. Node Exporter
D. Chef Server
ANSWER: C

Which component is listed for exposing Windows system metrics to Prometheus?
A. Node Exporter
B. Terraform
C. Puppet manifest
D. Windows Exporter
ANSWER: D

What is the purpose of a Prometheus scrape configuration?
A. Define targets and settings from which metrics are collected
B. Create AWS billing invoices
C. Define user desktop backgrounds
D. Publish Git tags
ANSWER: A

Which query language is used by Prometheus for selecting and aggregating time-series data?
A. SQL only
B. PromQL
C. YAML
D. Bash
ANSWER: B

What does a Prometheus target represent in a typical scrape setup?
A. A user password
B. A Dockerfile instruction
C. An endpoint from which Prometheus collects metrics
D. A SAN disk block
ANSWER: C

A Linux server is missing from Prometheus metrics. What should be checked first?
A. The Git commit author only
B. The server wallpaper
C. The cloud service catalog title
D. Exporter availability, target address, connectivity, and scrape status
ANSWER: D

What is the primary purpose of the Prometheus Web UI in this syllabus context?
A. Query and visualize collected metrics
B. Create a VPC subnet
C. Build a C executable
D. Configure a ZFS pool
ANSWER: A

Why are labels useful in Prometheus metrics?
A. They encrypt all metric values
B. They identify dimensions that can be used to filter and aggregate series
C. They replace all timestamps
D. They automatically create VM snapshots
ANSWER: B

Prometheus reports a target as down. Which interpretation is most appropriate?
A. The host is certainly powered off
B. The metric proves the application has no bugs
C. The scrape did not succeed; investigate reachability and endpoint configuration
D. The target has automatically been deleted
ANSWER: C

What is the main goal of DevOps?
A. Separate development and operations permanently
B. Eliminate testing
C. Replace all applications with spreadsheets
D. Improve collaboration and delivery through shared practices and automation
ANSWER: D

What does CI commonly stand for in CI/CD?
A. Continuous Integration
B. Container Installation
C. Central Internet
D. Configuration Index
ANSWER: A

What does CD commonly refer to in a CI/CD pipeline?
A. Central Directory only
B. Continuous Delivery or Continuous Deployment, depending on context
C. Container Disk
D. Code Deletion
ANSWER: B

What is a Docker image?
A. A running process only
B. A physical network card
C. A packaged, read-only template used to create containers
D. A service-level agreement
ANSWER: C

What is a Docker container?
A. A Git branch
B. A cloud billing report
C. A hypervisor type
D. An isolated running or stopped instance created from an image
ANSWER: D

Which command lists running Docker containers?
A. docker ps
B. docker pull
C. docker images
D. docker build
ANSWER: A

Which command builds an image from a Dockerfile in the current directory?
A. docker ps -a
B. docker build -t myapp .
C. docker logs myapp
D. docker network ls

ANSWER: B

Which command downloads an image from a registry?
A. docker commit nginx
B. docker inspect --type network
C. docker pull nginx
D. docker volume ls
ANSWER: C

A container exits immediately after startup. Which command is useful for reviewing its output?
A. docker image prune
B. docker network create
C. docker tag only
D. docker logs <container>
ANSWER: D

Why is a repeatable CI/CD pipeline useful?
A. It standardizes build, test, and delivery steps and reduces manual variation
B. It removes the need for source control
C. It guarantees all code is secure without testing
D. It prevents rollback in every case
ANSWER: A

Which Docker feature allows a container to communicate with other containers and networks?
A. Docker tag
B. Docker networking
C. Git commit
D. Service catalog
ANSWER: B

What is a Docker volume commonly used for?
A. Changing the CPU architecture
B. Creating a Git branch
C. Persisting data independently of a container's writable layer
D. Resolving DNS for every host automatically
ANSWER: C

What is Docker Compose designed to do?
A. Replace all container images with virtual machines
B. Monitor only Windows CPU metrics
C. Manage ITIL service portfolios
D. Define and manage multi-container applications using a configuration file
ANSWER: D

Which command commonly starts services defined in a Compose file?
A. docker compose up
B. docker login --logout
C. docker image ls
D. git status
ANSWER: A

What does publishing a container port with a mapping such as -p 8000:80 mean?
A. Container port 8000 forwards to host port 80 only in all cases
B. Host port 8000 forwards to container port 80
C. Both ports are automatically encrypted
D. The image is pushed to Docker Hub
ANSWER: B

Which command gives an image a new repository name or tag without rebuilding its layers?
A. docker rm
B. docker exec
C. docker tag
D. docker volume rm
ANSWER: C

Which command uploads a tagged image to Docker Hub after authentication?
A. docker pull
B. docker logs
C. docker inspect

D. docker push
ANSWER: D

A database container is recreated and its data must remain. Which approach is most appropriate?
A. Store the data in a properly configured persistent volume
B. Rename the container every hour
C. Keep all data only in the disposable container layer
D. Use docker ps as a backup
ANSWER: A

A Compose application has a web service and a database service. What is a useful benefit of Compose?
A. It converts all services into one operating-system kernel
B. It defines services and their configuration together for repeatable deployment
C. It removes the need for network configuration
D. It guarantees zero downtime without design work
ANSWER: B

Which command shows Docker networks known to the local Docker engine?
A. docker volume inspect
B. docker tag
C. docker network ls
D. docker save
ANSWER: C

Which command starts an HTTPD container named web5 and maps host port 8000 to container port 80?
A. docker run --name web5 -p 80:8000 httpd
B. docker push web5 httpd
C. docker build web5 -p 8000:80
D. docker run -d --name web5 -p 8000:80 httpd
ANSWER: D

In the syllabus HTTPD practical, which file is copied into the container to serve a custom webpage?
A. index.html
B. terraform.tfstate
C. Docker.lock
D. prometheus.yml
ANSWER: A

Which command copies a local index.html file into the web root of a running HTTPD container?
A. docker push index.html web5
B. docker cp index.html web5:/usr/local/apache2/htdocs/index.html
C. docker pull index.html
D. git clone web5:/index.html
ANSWER: B

After modifying the local index.html, what must be done to update the copied file inside the running container?
A. Run docker images only
B. Rename the Docker host
C. Copy the modified file into the container again or use an appropriate mounted content path
D. Delete the Docker engine
ANSWER: C

Which Dockerfile instruction selects the base image?
A. RUN
B. COPY
C. CMD
D. FROM
ANSWER: D

Which Dockerfile instruction copies files from the build context into the image?
A. COPY
B. WORKDIR
C. EXPOSE
D. ENTRYPOINT
ANSWER: A

What is the purpose of the CMD instruction in a Dockerfile?
A. Choose the registry password
B. Provide the default command or arguments when a container starts
C. Create an AWS subnet
D. Set the Git author globally
ANSWER: B

Why might a C program that requires interactive input need special handling in a container?
A. Docker cannot run compiled programs
B. C programs always require a GPU
C. The container must be run with suitable interactive input and terminal options
D. The Dockerfile must be named index.html
ANSWER: C

Which command can display the container's port mappings and other runtime details?
A. git push web5
B. docker compose buildx ls only
C. terraform output web5
D. docker inspect web5
ANSWER: D

What is the purpose of tagging an image before pushing it to a registry?
A. Give it a repository name and version/reference used for publishing
B. Convert the image into a VM snapshot
C. Encrypt every file inside it automatically
D. Create a Linux user
ANSWER: A

Which Dockerfile instruction selects a base image for a C-program container?
A. VOLUME
B. FROM
C. EXPOSE
D. LABEL
ANSWER: B

Which command creates a tar archive of a local Docker image?
A. docker logs -o cprog-back.tar image:tag
B. docker cp image:tag cprog-back.tar
C. docker save -o cprog-back.tar image:tag
D. docker network save image:tag
ANSWER: C

Which command loads a previously saved Docker image archive?
A. docker pull cprog-back.tar
B. docker run cprog-back.tar
C. docker volume import cprog-back.tar
D. docker load -i cprog-back.tar
ANSWER: D

After building a Docker image, which command can run it as a container?
A. docker run image:tag
B. docker images image:tag
C. docker tag image:tag
D. docker build image:tag
ANSWER: A

Which Dockerfile instruction can copy a compiled executable into an image?
A. EXPOSE
B. COPY
C. HEALTHCHECK only
D. STOPSIGNAL only
ANSWER: B

What should be verified when a containerized C program fails to execute?
A. Only the image's display name
B. Only the host desktop theme
C. Executable path, permissions, architecture, dependencies, and startup command
D. Only the GitHub repository description
ANSWER: C

Why should the base-machine version of the C program be tested before containerization?
A. It automatically validates every Docker network
B. It replaces all image testing
C. It guarantees the container needs no dependencies
D. It helps separate program defects from container configuration issues
ANSWER: D

Which option is commonly used with docker run to keep stdin open and allocate a terminal for an interactive program?
A. -it
B. --pull=never only
C. -d only
D. --rm only
ANSWER: A

A container exits because its configured command finishes. What does this usually indicate?
A. Docker always keeps every container alive
B. A container normally stops when its main process exits
C. The image has become a virtual machine
D. The registry deleted the host operating system
ANSWER: B

Why should a Docker image version be specified clearly when publishing a program image?
A. It changes the program's source code automatically
B. It disables image layers
C. It helps identify and retrieve the intended build
D. It replaces the need for testing
ANSWER: C

What is Git primarily used for?
A. Metric scraping
B. Cloud identity management
C. Container orchestration
D. Distributed version control
ANSWER: D

Which command configures the Git username for all repositories of the current user?
A. git config --global user.name "Name"
B. git push user.name
C. git clone --global user.name
D. git init user.name
ANSWER: A

Which command copies a remote repository to the local computer?
A. git status
B. git clone
C. git diff
D. git log
ANSWER: B

Which command shows the current working-tree and staging status?
A. git push
B. git remote add
C. git status
D. git tag
ANSWER: C

Which command stages a file named app.py?
A. git commit app.py
B. git clone app.py
C. git pull app.py
D. git add app.py
ANSWER: D

Which command records staged changes in the local repository history?
A. git commit -m "message"
B. git clone -m "message"
C. git status -m "message"

D. git remote -m "message"
ANSWER: A

Which command shows recent commit history?
A. git add
B. git log
C. git init
D. git config
ANSWER: B

What is the purpose of a Git branch?
A. Create a physical network interface
B. Allocate an AWS subnet
C. Develop work on a separate line of history
D. Store a Prometheus metric
ANSWER: C

A developer edited a file but has not staged it. Which command can show unstaged changes?
A. git push
B. git clone
C. git remote -v
D. git diff
ANSWER: D

Which sequence best represents a simple local Git workflow?
A. git add, then git commit
B. git clone, then disable version control
C. git push, then delete the repository
D. git status, then automatically publish all files
ANSWER: A

What is GitHub commonly used for in a Git workflow?
A. Providing HVAC control
B. Hosting remote Git repositories and supporting collaboration
C. Replacing all local commits
D. Collecting operating-system metrics only
ANSWER: B

Which command publishes local commits to a configured remote repository?
A. git add
B. git status
C. git push
D. git diff
ANSWER: C

Which command lists configured remotes and their URLs?
A. git branch -d
B. git log --oneline
C. git init
D. git remote -v
ANSWER: D

What does a successful git clone normally create?
A. A local copy of a repository with its history and remote configuration
B. A cloud IAM user
C. A running Docker container
D. A Kubernetes worker node
ANSWER: A

Why configure user.name and user.email in Git?
A. They set the GitHub password automatically
B. They identify the author metadata recorded in commits
C. They encrypt all commits by default
D. They define Docker port mappings
ANSWER: B

A developer committed changes locally but GitHub does not show them. Which step may still be required?
A. Run git status forever

B. Delete the local branch
C. Push the commits to the remote
D. Reinstall the operating system
ANSWER: C

What is a remote repository?
A. A file that exists only in RAM
B. A container's writable layer
C. A local shell alias
D. A repository hosted at another location, such as GitHub
ANSWER: D

Which practice is safest for authentication to a remote repository?
A. Use supported secure authentication and protect credentials
B. Share a password in a public issue
C. Commit personal access tokens into public code
D. Disable all access controls
ANSWER: A

What is a pull request commonly used for on GitHub?
A. Allocate an IP address
B. Propose and review changes before merging them
C. Start a SAN volume
D. Export a Docker image to tar
ANSWER: B

Before pushing, a developer wants to see which files are staged and modified. Which command should be used?
A. git push --force by default
B. git remote remove origin
C. git status
D. git init --bare
ANSWER: C

What is Jenkins primarily used for in DevOps?
A. Managing data center cooling
B. Replacing Git repositories
C. Providing block storage
D. Automating build, test, and delivery workflows
ANSWER: D

What is a Jenkins job or project?
A. A configured unit of work Jenkins can execute
B. A cloud subnet
C. A SAN disk partition
D. A Linux user group
ANSWER: A

What is a Jenkins pipeline?
A. A physical cable tray
B. A defined sequence of stages for automating delivery steps
C. A DNS zone
D. A virtual machine snapshot
ANSWER: B

Which file commonly defines a Pipeline as Code in a repository?
A. index.html only
B. terraform.tfstate only
C. Jenkinsfile
D. hosts.allow only
ANSWER: C

Why integrate Jenkins with GitHub?
A. To replace source control with manual email
B. To disable build logs
C. To provide physical access to a data center
D. To trigger or coordinate automation based on source-code changes
ANSWER: D

What is a build stage in a Jenkins pipeline used for?
A. Grouping related automation steps such as build or test
B. Assigning service provider type
C. Creating a new WAN circuit
D. Changing HVAC filters
ANSWER: A

A Jenkins build fails. Which information is a useful first diagnostic source?
A. The repository's logo only
B. The console output and stage logs
C. The server's rack color
D. The Docker Hub profile picture
ANSWER: B

What is the benefit of automating tests in Jenkins?
A. It guarantees software contains no defects
B. It eliminates code review in all cases
C. It provides repeatable feedback on changes
D. It makes version control unnecessary
ANSWER: C

When Jenkins builds a Docker image, what is commonly needed on the build agent?
A. A physical SAN controller for every image
B. A Windows Exporter configuration only
C. A service catalog entry only
D. Access to a compatible Docker engine or build capability
ANSWER: D

Which practice helps protect secrets used by a Jenkins pipeline?
A. Use controlled credentials storage and restrict access
B. Print secret values in every build log
C. Hard-code passwords in a public Jenkinsfile
D. Grant every job unrestricted administrator credentials
ANSWER: A

Which command builds a Docker image and tags it as webapp:v1?
A. docker push webapp:v1 .
B. docker build -t webapp:v1 .
C. docker run -t webapp:v1 .
D. git commit -t webapp:v1
ANSWER: B

Which command pushes a tagged image to a configured registry?
A. docker ps repository/image:tag
B. docker logs repository/image:tag
C. docker push repository/image:tag
D. docker rm repository/image:tag
ANSWER: C

Which command exports an image to the required archive file cprog-back.tar?
A. docker cp image:tag cprog-back.tar
B. docker logs image:tag > cprog-back.tar
C. docker network inspect image:tag
D. docker save -o cprog-back.tar image:tag
ANSWER: D

In a Jenkins pipeline using GitHub, what is the repository integration commonly used for?
A. Retrieving source code and detecting or responding to changes
B. Replacing every build stage
C. Providing server cooling
D. Creating physical network ports
ANSWER: A

What is the role of NGINX or HTTPD in the syllabus web pipeline exercise?
A. Compile Terraform configuration
B. Serve web content from a container image
C. Collect Windows metrics
D. Manage Chef cookbooks
ANSWER: B

A pipeline should publish a new image only after tests pass. How should the pipeline be arranged?
A. Push first and skip tests
B. Remove the test stage
C. Run tests before the image-publishing stage
D. Run only a Git status command
ANSWER: C

Why use versioned tags for Docker images in a CI/CD pipeline?
A. To guarantee the image is vulnerability-free
B. To remove the need for a registry
C. To convert an image into a Git branch
D. To identify specific builds and support controlled deployments
ANSWER: D

Jenkins cannot connect to the Docker daemon. Which area should be checked?
A. Daemon availability, socket permissions, and agent configuration
B. The application's page title only
C. Git commit message punctuation only
D. The cloud service catalog
ANSWER: A

What does a pipeline artifact commonly represent?
A. A physical cooling fan
B. A file or package produced by a build for later use
C. An IAM role name only
D. A DNS cache entry
ANSWER: B

Why should credentials used to push to Docker Hub be stored securely in Jenkins?
A. To make the repository public automatically
B. To bypass authentication
C. To limit exposure and control access to registry operations
D. To remove image tags
ANSWER: C

Which AWS service provides resizable virtual servers?
A. Amazon S3
B. AWS Lambda
C. Amazon Route 53 only
D. Amazon EC2
ANSWER: D

Which AWS service runs code in response to events without requiring users to manage a continuously running server?
A. AWS Lambda
B. Amazon VPC
C. Amazon S3
D. Amazon EBS only
ANSWER: A

Which AWS service provides object storage?
A. Amazon EC2
B. Amazon S3
C. AWS Lambda
D. Amazon VPC
ANSWER: B

What does Amazon VPC provide?
A. A source-code version history
B. A Docker image build file
C. A logically isolated virtual network for AWS resources
D. A physical HVAC system
ANSWER: C

What is an EC2 instance?
A. An object storage bucket
B. A Git commit
C. A Prometheus query

D. A virtual compute server in AWS
ANSWER: D

Which AWS resource is used to store objects such as files and backups?
A. S3 bucket
B. Lambda execution role only
C. EC2 security group only
D. VPC route table only
ANSWER: A

Which principle should guide permissions assigned to an AWS identity?
A. Unrestricted access by default
B. Least privilege
C. Shared root credentials
D. Public exposure of all keys
ANSWER: B

What is the purpose of a VPC subnet?
A. Store Docker image layers
B. Track Git commits
C. Divide a VPC address range into smaller network segments
D. Define Jenkins build stages
ANSWER: C

A workload needs compute capacity for a long-running web server. Which listed service is the direct fit?
A. Amazon S3
B. AWS Lambda only
C. CloudWatch metric label
D. Amazon EC2
ANSWER: D

Which practice is appropriate for AWS credentials used by automation?
A. Use controlled credentials or roles and avoid hard-coding secrets
B. Send keys in plain text to all users
C. Commit access keys into a public repository
D. Disable all permission boundaries
ANSWER: A

What is the CIDR network address range specified for the ditiss-lab VPC?
A. 172.20.5.0/24
B. 172.20.0.0/16
C. 172.20.10.0/24
D. 192.168.1.0/24
ANSWER: B

Which CIDR range is specified for the public subnet in the syllabus lab?
A. 172.20.0.0/16
B. 172.20.10.0/24
C. 172.20.5.0/24
D. 10.0.0.0/8
ANSWER: C

Which CIDR range is specified for the private subnet in the syllabus lab?
A. 172.20.5.0/24
B. 172.20.0.0/16
C. 192.168.0.0/16
D. 172.20.10.0/24
ANSWER: D

Which subnet is intended to host the instance directly reachable through the lab's public connectivity setup?
A. Public subnet
B. S3 bucket
C. Private subnet
D. Lambda layer
ANSWER: A

Which service is to be installed on the private EC2 instance in the lab?

A. Prometheus only
B. HTTPD
C. Chef Server only
D. GitHub Desktop
ANSWER: B

Which command is specified as a way to test the web service from the public instance?
A. git clone
B. terraform fmt
C. curl
D. docker save
ANSWER: C

For an EC2 instance to be reachable from the internet, which combination is generally relevant?
A. Only a private IP with no route
B. Only an S3 bucket name
C. Only a Git branch
D. Public addressing or suitable routing, route table, and permitted security-group traffic
ANSWER: D

A public instance cannot curl the private instance's web server. Which check is most relevant?
A. Private instance service status, security groups, routes, and target address
B. The Docker image tag unrelated to the service
C. The public instance's wallpaper
D. The Jenkins job description
ANSWER: A

Why should a private subnet generally not provide direct inbound internet access to its instances?
A. It makes IP addressing impossible
B. It reduces direct exposure; access can be mediated through controlled paths
C. It prevents internal communication by definition
D. It automatically disables HTTPD
ANSWER: B

Which statement about the specified subnet ranges is correct?
A. The 172.20.5.0/24 subnet is outside 172.20.0.0/16
B. The 172.20.10.0/24 subnet overlaps the public subnet exactly
C. Both /24 subnets are contained within the 172.20.0.0/16 VPC range
D. A /16 range is smaller than a /24 range
ANSWER: C

What is container orchestration designed to manage?
A. Only Git author names
B. Only physical cooling
C. Only object-storage billing
D. Deployment, scheduling, scaling, and lifecycle of containers
ANSWER: D

Which orchestration platform is explicitly named alongside Docker Swarm in the syllabus?
A. Kubernetes
B. Prometheus
C. Chef
D. FreeNAS
ANSWER: A

Which Docker Swarm object defines a desired state for replicated tasks?
A. Git commit
B. Service
C. ZFS pool
D. Service Design Package
ANSWER: B

What does replica count represent for a replicated service?
A. The number of Git remotes
B. The size of a CIDR prefix
C. The desired number of task or pod instances
D. The number of Dockerfiles in a directory
ANSWER: C

What is a microservice architecture based on?
A. One mandatory physical server per user
B. A single shared executable for every unrelated function
C. A network with no APIs
D. Deployable services organized around relatively focused capabilities
ANSWER: D

In the syllabus lab, what is the initial Docker Swarm replica count?
A. 1
B. 10
C. 2
D. 100
ANSWER: A

Which command is commonly used to create or update a Docker Swarm service's replica count?
A. docker image prune
B. docker service scale
C. git branch
D. terraform validate
ANSWER: B

What is a worker node's general role in a container cluster?
A. Act only as a Git remote
B. Store service-level agreements
C. Run scheduled workloads assigned by the orchestration system
D. Replace all control-plane functions in every architecture
ANSWER: C

Why is a container image registry used in the orchestration lab?
A. To allocate subnet CIDRs
B. To monitor HVAC
C. To replace the cluster scheduler
D. To store and distribute images for deployment
ANSWER: D

What should be checked if a service fails to scale to the requested replicas?
A. Cluster capacity, service configuration, scheduling events, and node health
B. Only the Git username
C. Only the image's logo
D. Only the file extension of a report
ANSWER: A

In the Docker Swarm lab, what replica count is required after the first scale-up?
A. 1
B. 10
C. 2
D. 20
ANSWER: B

In the Docker Swarm lab, what replica count is required after scaling down?
A. 1
B. 10
C. 2
D. 0
ANSWER: C

After modifying index.html and pushing a new image, what is needed to serve the new content through a service?
A. Only rename the local source file
B. Restart GitHub
C. Change the VPC CIDR
D. Update the service to use the new image version
ANSWER: D

What is a Kubernetes Pod?
A. The smallest deployable unit that can contain one or more containers
B. A Terraform state backend
C. A Docker Hub account
D. A physical network switch

ANSWER: A

What is the primary role of the Kubernetes control plane?
A. Serve every application request directly by definition
B. Manage cluster state and coordinate workload scheduling and control
C. Replace container images
D. Store all source code as Git commits
ANSWER: B

What is a Kubernetes worker node responsible for?
A. Creating all business strategy documents
B. Replacing the image registry
C. Running pods and node-level services
D. Managing ITIL financial charging
ANSWER: C

Which Kubernetes resource commonly manages a desired number of replicated pods?
A. ConfigMap only
B. Namespace only
C. PersistentVolume only
D. Deployment
ANSWER: D

A new image was pushed but Kubernetes continues running the old version. What should be checked?
A. The workload image reference and rollout status
B. Only the host wallpaper
C. Only the cluster name's spelling
D. Only the Git user email
ANSWER: A

Why should cluster nodes have compatible networking and connectivity configured?
A. It ensures no application needs authentication
B. The control plane, nodes, and workloads need communication for cluster operation
C. It removes the need for images
D. It guarantees all nodes have identical hardware
ANSWER: B

What is a sensible first response when a Kubernetes pod remains Pending?
A. Delete every namespace immediately
B. Rebuild all images without checking events
C. Inspect pod events, resource requests, scheduling constraints, and node capacity
D. Change the Docker Hub password only
ANSWER: C

What is Ansible primarily used for?
A. Providing block storage
B. Collecting metrics only
C. Replacing all source control
D. Automating configuration, deployment, and orchestration tasks
ANSWER: D

Which format is commonly used to write Ansible playbooks?
A. YAML
B. Docker image layers
C. PromQL
D. CIDR notation only
ANSWER: A

What is an Ansible inventory?
A. A list of Docker image layers
B. A list or description of managed hosts and groups
C. A service-level agreement
D. A VM snapshot
ANSWER: B

Which authentication method is specified in the syllabus lab for passwordless target access?
A. Publicly shared root password
B. Anonymous FTP
C. SSH key-based authentication

D. Unauthenticated HTTP
ANSWER: C

What does the Ansible ping module check?
A. ICMP reachability alone in every case
B. Whether a website's content is correct
C. Whether Docker Hub is public
D. Basic Ansible connectivity and Python execution on managed hosts
ANSWER: D

Which command is the syllabus's stated inventory connectivity check?
A. ansible -m ping all
B. docker build -m ping
C. git push all
D. terraform plan all
ANSWER: A

What is an Ansible playbook?
A. A Docker registry
B. A YAML file describing plays, hosts, and tasks
C. A network monitoring dashboard
D. A cloud subnet
ANSWER: B

What is a major benefit of Ansible roles?
A. Replacing SSH with physical access
B. Creating a new operating-system kernel
C. Organizing reusable automation content into a standard structure
D. Disabling variables
ANSWER: C

Why is idempotence useful in configuration automation?
A. Every run must create duplicates
B. It prevents all errors automatically
C. It removes the need for inventory
D. Repeated runs converge toward the same desired state without unnecessary changes
ANSWER: D

A playbook cannot connect to a target host. Which items should be checked first?
A. Inventory, hostname resolution, SSH access, credentials, and privilege escalation
B. Only the target's wallpaper
C. Only the YAML file's font
D. Only the Docker image registry description
ANSWER: A

Which Ansible module is commonly used to install packages on supported Linux systems?
A. copy only
B. package
C. debug only
D. assert only
ANSWER: B

Which Ansible module can ensure a service is running and enabled?
A. template only
B. set_fact only
C. service
D. fail only
ANSWER: C

Which file commonly lists managed hosts in a simple static Ansible inventory?
A. Jenkinsfile only
B. Dockerfile only
C. terraform.tfstate only
D. An inventory file such as hosts
ANSWER: D

What is the purpose of the become mechanism in Ansible?
A. Run tasks with privilege escalation when configured
B. Create Docker volumes

C. Clone a Git repository automatically in every play
D. Expose Prometheus metrics
ANSWER: A

Which Ansible module is suitable for rendering a configuration file from a Jinja2 template?
A. ping
B. template
C. meta only
D. wait_for only
ANSWER: B

Why should an Ansible role be used for reusable database deployment tasks?
A. It forces the database to run without authentication
B. It replaces all target inventories
C. It separates related defaults, variables, tasks, handlers, and files into a reusable structure
D. It stores database passwords publicly
ANSWER: C

What is a handler commonly used for in Ansible?
A. Create a VPC subnet automatically in every play
B. Store Git commit history
C. Replace the inventory file
D. Run an action such as restarting a service when notified by a task
ANSWER: D

A playbook reports success but the web server is not reachable. What should be verified?
A. Service state, listening port, firewall rules, and webpage access
B. Only the GitHub avatar
C. Only the playbook filename
D. Only the cloud account display name
ANSWER: A

Which practice is appropriate for secrets used by Ansible?
A. Store them in a public inventory without encryption
B. Protect them with an approved secret-management method such as Ansible Vault
C. Print them in every task output
D. Use the same unrestricted password everywhere
ANSWER: B

Why is a multi-host inventory useful?
A. It replaces the need for SSH connectivity
B. It makes every host a controller
C. It lets automation target groups of managed systems
D. It automatically creates physical servers
ANSWER: C

What does Infrastructure as Code (IaC) mean?
A. Configuring all systems only through manual clicks
B. Writing application user manuals only
C. Storing every metric in a spreadsheet
D. Managing infrastructure through versionable machine-readable definitions
ANSWER: D

Which command initializes a Terraform working directory and its providers or modules?
A. terraform init
B. terraform destroy
C. terraform apply
D. terraform output
ANSWER: A

Which command previews proposed infrastructure changes?
A. terraform fmt
B. terraform plan
C. terraform output
D. terraform providers lock only
ANSWER: B

Which command applies a Terraform configuration to create or modify resources?
A. terraform validate

B. terraform version
C. terraform apply
D. terraform state list only
ANSWER: C

Which Terraform block declares an input value that can be supplied or overridden?
A. output
B. provider only
C. terraform only
D. variable
ANSWER: D

Which Terraform block exposes a value after configuration is applied?
A. output
B. locals only
C. variable
D. required_providers only
ANSWER: A

What is the purpose of a Terraform provider?
A. Store HTML pages inside a Docker container
B. Allow Terraform to interact with an API or platform such as AWS
C. Collect Linux metrics
D. Configure Git commit authors
ANSWER: B

Which command checks Terraform configuration for syntax and internal consistency errors?
A. terraform push
B. terraform logs
C. terraform validate
D. terraform clone
ANSWER: C

Why should AWS credentials used by Terraform be configured securely?
A. Credentials automatically determine the CIDR of every VPC
B. Credentials replace resource configuration
C. Credentials make state files unnecessary
D. Terraform needs authorized API access to create and manage AWS resources
ANSWER: D

What is a key benefit of using variables in Terraform configurations?
A. Reuse the configuration with different input values
B. Remove the need for providers
C. Prevent all infrastructure changes
D. Guarantee that resources never fail
ANSWER: A

What does Terraform state track?
A. Only the GitHub login page
B. The relationship between Terraform-managed resource instances and configuration
C. Only current CPU utilization
D. Only a Docker container's console output
ANSWER: B

What is a Terraform module?
A. A Prometheus exporter
B. A physical storage controller
C. A reusable collection of Terraform configuration files
D. A Jenkins build log
ANSWER: C

Why use a remote Terraform state backend?
A. Make state unnecessary
B. Expose cloud credentials publicly
C. Run Terraform without configuration
D. Share state through a controlled remote location for collaboration
ANSWER: D

Which AWS service is given as an example for a remote Terraform state backend in the syllabus?

A. Amazon S3
B. Amazon EC2 only
C. AWS Lambda
D. Amazon Route 53 only
ANSWER: A

Why should concurrent Terraform operations against shared state be coordinated?
A. To ensure every user overwrites the state
B. To reduce conflicting updates and state inconsistency
C. To disable provider authentication
D. To eliminate all infrastructure drift automatically
ANSWER: B

What is the purpose of a Terraform security group resource in an EC2 module?
A. Store application source code
B. Collect Prometheus time series
C. Define permitted network traffic to or from supported resources
D. Define a GitHub pull request
ANSWER: C

Which configuration elements are explicitly part of the syllabus's VPC provisioning exercise?
A. Only a Docker volume and Git branch
B. Only a Jenkins job and Chef recipe
C. Only an ITIL service catalog
D. VPC, subnet, and EC2 instance
ANSWER: D

What should be done before applying a Terraform change to production infrastructure?
A. Review the plan and confirm the intended resource changes
B. Publish state and credentials publicly
C. Run destroy without inspection
D. Delete the configuration files
ANSWER: A

Why should Terraform state files be protected?
A. They contain only public documentation
B. They can contain sensitive values and important resource metadata
C. They are identical to a Git README
D. They are never needed after creation
ANSWER: B

A Terraform apply proposes replacing an important EC2 instance unexpectedly. What is the safest next step?
A. Approve immediately without review
B. Delete the remote state file
C. Inspect the plan and configuration/state causes before approving changes
D. Disable all provider authentication
ANSWER: C